Econometrics (25117): Midterm 1, 2026–2027
Concise answer key
Monday, 9 November 2026

MCQs: 2.5 points each (20 points); Questions 9–10: 10 points each; Questions 11–15: 12 points each. Total: 100 points. Full explanations and lecture references: detailed answer key.

Multiple-choice questions

Question	1	2	3	4	5	6	7	8
Answer	b	c	b	a	d	b	d	c

Long-answer questions

Question 9: joint probabilities and dependence (10 points)

(a) $\Pr(D = 1) = 0.24 + 0.36 = 0.60$; $\Pr(Y = 1) = 0.04 + 0.36 = 0.40$.

(b) $\mathrm{E}[Y \mid D = 0] = 0.04/0.40 = 0.10$; $\mathrm{E}[Y \mid D = 1] = 0.36/0.60 = 0.60$. $\mathrm{E}[Y] = 0.10(0.40) + 0.60(0.60) = 0.40$.

(c) $\mathrm{E}[DY] = 0.36$ and $\mathrm{Var}(D) = \mathrm{Var}(Y) = 0.24$, so

$$\mathrm{Cov}(D, Y) = 0.36 - (0.60)(0.40) = 0.12, \qquad \mathrm{Corr}(D, Y) = \frac{0.12}{\sqrt{0.24 \cdot 0.24}} = 0.50.$$

(d) Not independent: $\Pr(D = 1, Y = 1) = 0.36 \neq \Pr(D = 1)\Pr(Y = 1) = 0.24$.

Question 10: comparing two means (10 points)

(a) $\widehat{\Delta} = 74 - 68 = 6$ score points; $SE(\widehat{\Delta}) = \sqrt{12^2/100 + 16^2/100} = 2$ score points.

(b) $H_0 : \Delta = 2$ versus $H_1 : \Delta \neq 2$; $t_{\text{obs}} = (6 - 2)/2 = 2$.

(c) At 5%: $|2| > 1.96$, reject H_0. At 1%: $|2| < 2.576$, do not reject H_0.

(d) $CI_{95\%} = 6 \pm 1.96(2) = [2.08, 9.92]$ score points. It excludes 2, consistently with the two-sided 5% rejection.

(e) The planned half-width satisfies

$$1.96\sqrt{\frac{12^2 + 16^2}{m}} = \frac{39.2}{\sqrt{m}} \leq 2 \quad \Longleftrightarrow \quad m \geq 384.16.$$

$m_{\min} = 385$ students per group; total $2m_{\min} = 770$ students.

Question 11: a small library population (12 points)

(a) $\Pr(Y = 0) = 1/8$, $\Pr(Y = 2) = 3/4$ and $\Pr(Y = 4) = 1/8$;

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ 1/8, & 0 \leq y < 2, \\ 7/8, & 2 \leq y < 4, \\ 1, & y \geq 4. \end{cases}$$

(b) $\mathrm{E}[Y] = 0(1/8) + 2(3/4) + 4(1/8) = 2$ books; $\mathrm{E}[Y^2] = 0^2(1/8) + 2^2(3/4) + 4^2(1/8) = 5$ books squared; $\mathrm{Var}(Y) = 5 - 2^2 = 1$ book squared.

(c) $\Pr(G = \text{A}) = \Pr(G = \text{B}) = 1/2$. $\text{E}[Y \mid G = \text{A}] = [0(1) + 2(2) + 4(1)]/4 = 2$ books and $\text{E}[Y \mid G = \text{B}] = 2(4)/4 = 2$ books. $\text{E}[Y] = 2(1/2) + 2(1/2) = 2$ books.

(d) Not independent: $\Pr(Y = 2 \mid G = \text{B}) = 1 \neq 3/4 = \Pr(Y = 2)$. Equal conditional means do not imply independence.

Question 12: from sample summaries to inference (12 points)

(a)

$$\bar{y} = 5200/100 = 52 \text{ points}, \qquad S_{YY} = 280300 - 100(52)^2 = 9900 \text{ points}^2.$$

$$s_Y^2 = 9900/99 = 100 \text{ points}^2, \qquad s_Y = 10 \text{ points}, \qquad SE(\bar{Y}) = 10/\sqrt{100} = 1 \text{ point}.$$

(b) $H_0 : \mu_Y = 50$ versus $H_1 : \mu_Y \neq 50$; $t_{\text{obs}} = (52 - 50)/1 = 2$. At 5%: $|2| > 1.96$, reject H_0. At 1%: $|2| < 2.576$, do not reject H_0.

(c) $CI_{95\%} = 52 \pm 1.96(1) = [50.04, 53.96]$ points. It excludes 50, consistently with the two-sided 5% rejection.

(d)

$$n_{\min} = \left\lceil \left(\frac{1.96(10)}{0.98} \right)^2 \right\rceil = 400 = 4(100).$$

With the planning SD unchanged, the planned SE halves from 1 to 0.5 points and the planned half-width halves from 1.96 to 0.98 points.

Question 13: simple OLS (12 points)

(a) Minimize $Q(b_0, b_1) = \sum_i (y_i - b_0 - b_1 x_i)^2$. The first-order conditions are

$$\frac{\partial Q}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i) = 0, \qquad \frac{\partial Q}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i) = 0.$$

The first gives $\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}$. Substituting into the second and centering gives

$$0 = \sum_i (x_i - \bar{x})(y_i - \bar{y}) - \widehat{\beta}_1 \sum_i (x_i - \bar{x})^2, \qquad \widehat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}.$$

The solution is unique when $\sum_i (x_i - \bar{x})^2 > 0$, so the x_i are not all equal.

(b) $\widehat{\beta}_1 = 50/100 = 0.5$, $\widehat{\beta}_0 = 70 - 0.5(4) = 68$; $\widehat{y}_i = 68 + 0.5 x_i$. One additional weekly study hour is associated with 0.5 additional predicted score points.

(c) $\widehat{y}_i = 68 + 0.5(6) = 71$; $\widehat{u}_i = 72 - 71 = 1$ point.

(d) $ESS = 0.5^2(100) = 25$, $SSR = 425 - 25 = 400$; $R^2 = 25/425 = 1/17$, $SER = \sqrt{400/(102 - 2)} = 2$ points.

(e) $r_{XY} = 50/\sqrt{100(425)} = 1/\sqrt{17}$; $r_{XY}^2 = 1/17 = R^2$.

(f) $\widehat{y}_i = 68 + M_i/120$: intercept 68, slope 1/120 point per minute; $R^2 = 1/17$ and $SER = 2$ points.

Question 14: one regressor without an intercept (12 points)

(a) Minimize $Q(b_1) = \sum_i (y_i - b_1 x_i)^2$. Then

$$Q'(b_1) = -2 \sum_i x_i (y_i - b_1 x_i) = 0 \quad \Longrightarrow \quad \widehat{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2}.$$

Uniqueness requires $\sum_i x_i^2 > 0$ (at least one nonzero x_i); then $Q''(b_1) = 2 \sum_i x_i^2 > 0$.

(b) The raw sums and estimate are

$$\sum_i x_i^2 = 100 + 102(4)^2 = 1732, \qquad \sum_i x_i y_i = 50 + 102(4)(70) = 28610,$$

$$\widehat{\beta}_1 = \frac{28610}{1732} = \frac{14305}{866}, \qquad \widehat{y}_i = \frac{14305}{866} x_i.$$

One additional weekly study hour adds $14305/866 \approx 16.5185$ predicted score points in this imposed line through the origin.

(c) $\widehat{y}_i = 6(14305/866) = 42915/433 \approx 99.11$ points; $\widehat{u}_i = 72 - 42915/433 = -11739/433 \approx -27.11$ points.

(d)

$$\sum_i x_i \widehat{u}_i = 28610 - \frac{14305}{866}(1732) = 0, \qquad \sum_i \widehat{u}_i = 7140 - \frac{14305}{866}(408) = \frac{173400}{433}.$$

The ordinary sum need not be zero: without an intercept, there is no intercept first-order condition imposing it.

Question 15: simple OLS from five observations (12 points)

(a) $\bar{x} = 3$, $\bar{y} = 7$, $S_{XX} = 10$, $S_{XY} = 20$; $\widehat{\beta}_1 = 20/10 = 2$, $\widehat{\beta}_0 = 7 - 2(3) = 1$; $\widehat{y}_i = 1 + 2x_i$. One additional weekly study hour is associated with two additional predicted quiz points.

(b) Fitted values and residuals:

i	1	2	3	4	5
$\widehat{y}_i$	3	5	7	9	11
$\widehat{u}_i$	1	−1	0	−1	1
$x_i \widehat{u}_i$	1	−2	0	−4	5

$\sum_i \widehat{u}_i = 1 - 1 + 0 - 1 + 1 = 0$; $\sum_i x_i \widehat{u}_i = 1 - 2 + 0 - 4 + 5 = 0$.

(c) $TSS = 44$, $ESS = 40$, $SSR = 4$; $44 = 40 + 4$.

$$R^2 = \frac{40}{44} = \frac{10}{11}, \qquad SER = \sqrt{\frac{4}{5-2}} = \frac{2}{\sqrt{3}} \text{ points.}$$

(d) $\widehat{y}_i = 1 + M_i/30$: intercept 1, slope 1/30 point per minute; $R^2 = 10/11$, $SER = 2/\sqrt{3}$ points. Fitted scores and residuals are unchanged, so both fit statistics are unchanged.